RACE Regular Analysis Through Continuous Exercise

Trigonometric Equations: RACE-01

Single Correct Question

1. tan $(p\pi/4)$ = cot $(q\pi/4)$ if:

(A) $p + q = 0$ (B) $p + q = 2n + 1$ (C) $p + q = 2n$ (D) $p + q = 2(2n + 1)$

MTE133

2. The general solution of the equation, $\frac{1-\sin x+.....+(-1)^{n-1}\sin^{n-1}x+...\infty}{1+\sin x+....+\sin^{n-1}x+...\infty} = \frac{1-\cos 2x}{1+\cos 2x}$ is

(A) $(-1)^n\pi/3 + n\pi$ (B) $(-1)^n\pi/6 + n\pi$

(C) $(-1)^{n+1}\pi/6 + n\pi$ (D) $(-1)^{n-1}\pi/3 + n\pi$

MTE134

Multiple Type Question

3. cos $15x$ = sin $5x$ if

(A) $x = -\frac{\pi}{20} + \frac{n\pi}{5}, n \in I$ (B) $x = \frac{\pi}{40} + \frac{n\pi}{10}, n \in I$

(C) $x = \frac{3\pi}{20} + \frac{n\pi}{5}, n \in I$ (D) $x = -\frac{3\pi}{40} + \frac{n\pi}{10}, n \in I$

MTE135

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Subjective Type Question

4. Find general solutions of the following equations:

(a) $\sin\theta = \frac{1}{2}$ (b) $\cos\left(\frac{3\theta}{2}\right) = 0$ (c) $\tan\left(\frac{3\theta}{4}\right) = 0$ (d) $\cos^2 2\theta = 1$

(e) $\sqrt{3}\sec 2\theta = 2$ (f) $\text{cosec}\left(\frac{\theta}{2}\right) = -1$

MTE136

5. What are the most general values θ, which satisfy the equations :

(i) $\sin\theta = \frac{1}{\sqrt{2}}$ (ii) $\tan(x-1) = \sqrt{3}$ (iii) $\tan\theta = -1$

(iv) $\text{cosec}\theta = \frac{2}{\sqrt{3}}$ (v) $2\cot^2\theta = \text{cosec}^2\theta$

MTE137

6. Solve

(i) $\sin 9\theta = \sin\theta$ (ii) $\cot\theta + \tan\theta = 2\text{cosec}\theta$ (iii) $\sin 2\theta = \cos 3\theta$

MTE138

(iv) $\cot\theta = \tan 8\theta$ (v) $\cot\theta - \tan\theta = 2$ (vi) $\text{cosec}\theta = \cot\theta + \sqrt{3}$

MTE139

(vii) $\tan 2\theta \tan\theta = 1$ (viii) $\tan\theta + \tan 2\theta + \sqrt{3}\tan\theta\tan 2\theta = \sqrt{3}$

MTE140

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7. Solve the following equations:

(i) $3\sin x + 2\cos^2 x = 0$ (ii) $\sec^2 2\alpha = 1 - \tan^2 \alpha$

MTE141

(iii) $7\cos^2\theta + 3\sin^2\theta = 4$ (iv) $4\cos\theta - 3\sec\theta = \tan\theta$

MTE142

8. Solve $\cot^2\theta + 3\operatorname{cosec}\theta + 3 = 0$

MTE143

9. Solve the equation: $2\sin^2\theta + \sin^2 2\theta = 2$ for $\theta \in (-\pi, \pi)$.

MTE144

10. Solve $\cos^3 x + \cos^2 x - 4\cos^2 \frac{x}{2} = 0$

MTE145

ANSWER KEY

Question	1	2	3
Answer	D	B	A,B,C,D

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4. **(a)** $\theta = n\pi + (-1)^n \frac{\pi}{6}, n \in I$; **(b)** $\theta = (2n+1)\frac{\pi}{3}, n \in I$; **(c)** $\theta = \frac{4n\pi}{3}, n \in I$;

(d) $\theta = \frac{n\pi}{2}, n \in I$ **(e)** $\theta = n\pi \pm \frac{\pi}{12}, n \in I$; **(f)** $\theta = 2n\pi + (-1)^{n+1}\pi, n \in I$

5. **(i)** $n\pi + (-1)^n \frac{\pi}{4}, n \in I$; **(ii)** $n\pi + \frac{\pi}{3} + 1, n \in I$; **(iii)** $n\pi - \frac{\pi}{4}, n \in I$; **(iv)** $n\pi + (-1)^n \frac{\pi}{3}, n \in I$

(v) $n\pi \pm \frac{\pi}{4}, n \in I$

6. **(i)** $\frac{m\pi}{4}, m \in I$ or $\frac{(2m+1)\pi}{10}, m \in I$; **(ii)** $2n\pi \pm \frac{\pi}{3}, n \in I$

(iii) $\left(2n + \frac{1}{2}\right)\frac{\pi}{5}, n \in I$ or $2n\pi - \frac{\pi}{2}, n \in I$; **(iv)** $\left(n + \frac{1}{2}\right)\frac{\pi}{9}, n \in I$ **(v)** $\left(n + \frac{1}{4}\right)\frac{\pi}{2}, n \in I$

(vi) $2n\pi + \frac{2\pi}{3}, n \in I$; **(vii)** $n\pi \pm \frac{\pi}{6}$ **(viii)** $\left(n + \frac{1}{3}\right)\frac{\pi}{3}, n \in I$

7. **(i)** $x = n\pi + (-1)^{n+1}\frac{\pi}{6}, n \in I$; **(ii)** $\alpha = \frac{n\pi}{2}$ or $\alpha = \frac{k\pi}{2} + \frac{3\pi}{8}, n, k \in I$

(iii) $\theta = n\pi \pm \frac{\pi}{3}, n \in I$; **(iv)** $\theta = n\pi + (-1)^n \alpha$,

8. $2n\pi - \frac{\pi}{2}, n \in I$ or $n\pi + (-1)^{n+1}\frac{\pi}{6}, n \in I$ **9.** $\theta = \left\{-\frac{\pi}{4}, -\frac{3\pi}{4}, -\frac{\pi}{2}, \frac{\pi}{4}, \frac{3\pi}{4}, \frac{\pi}{2}\right\}$

10. $(2n+1)\pi, n \in I$

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RACE Regular Analysis Through Continuous Exercise

Trigonometric Equations: RACE-02

Single Correct Question

1. If sum of all possible values of $x \in (0, 2\pi)$ satisfying the equation

$2\cos x \cdot \operatorname{cosec} x - 4\cos x - \operatorname{cosec} x = -2$, is $\frac{k\pi}{4}$, then k is equal to

(A) 9 (B) 12

(C) 16 (D) 32

MTE146

Integer Type Question

2. If sum of all solutions of the equation $6\sin\frac{x}{2} = \sec\frac{x}{2}$ in $[-2\pi, 4\pi]$ is $k\pi$, then k is equal to

MTE147

Subjective Type Question

3. Solve : $\sin 2x + 5\sin x + 1 + 5\cos x = 0$

MTE148

4. Solve

(i) $\tan^2\theta - (1 + \sqrt{3})\tan\theta + \sqrt{3} = 0$

MTE149

(ii) $4\cos\theta - 3\sec\theta = 2\tan\theta$

MTE150

(iii) $\tan x \cdot \tan\left(x + \frac{\pi}{3}\right) \cdot \tan\left(x + \frac{2\pi}{3}\right) = \sqrt{3}$

MTE151

5. Solve the following equations :

(a) $\sin x + \sqrt{2} = \cos x$

(b) $\operatorname{cosec}\theta = 1 + \cot\theta$

MTE152

6. Solve : $8\sin x = \frac{\sqrt{3}}{\cos x} + \frac{1}{\sin x}$

MTE153

7. Solve :

(i) $\sqrt{3}\sin\theta - \cos\theta = \sqrt{2}$

MTE154

(ii) $5\sin\theta + 2\cos\theta = 5$

MTE155

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8. Solve for x: $\sin x + \sin 3x + \sin 5x = 0$.

MTE156

9. Solve : $\cos\theta + \sin\theta = \cos 2\theta + \sin 2\theta$

MTE157

10. Solve : $4\sin\theta \sin 2\theta \sin 4\theta = \sin 3\theta$.

MTE158

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ANSWER KEY

1. (B) **2.** 3 **3.** $n\pi - \frac{\pi}{4}, n \in I$

4. (i) $n\pi + \frac{\pi}{3}, n \in I$ or $n\pi + \frac{\pi}{4}, n \in I$

(ii) $n\pi + (-1)^n \frac{\pi}{10}, n \in I$ or $n\pi - (-1)^n \frac{3\pi}{10}, n \in I$

(iii) $x = \frac{n\pi}{3} - \frac{\pi}{9}, n \in I$

5. (a) $x = 2n\pi - \frac{\pi}{4}, n \in I$ (b) $2m\pi + \frac{\pi}{2}, m \in I$

6. $x = n\pi + \frac{\pi}{6}, n \in I, x = \frac{n\pi}{2} - \frac{\pi}{12}, n \in I$

7. (i) $n\pi + \frac{\pi}{6} + (-1)^n \frac{\pi}{4}, n \in I$; (ii) $2n\pi + \frac{\pi}{2}, n \in I$ or $2n\pi + 2\alpha$ where $\alpha = \tan^{-1}\frac{3}{7}, n \in I$

8. $x = \frac{n\pi}{3}, n \in I$ and $k\pi \pm \frac{\pi}{3}, k \in I$ **9.** $2n\pi, n \in I$ or $\frac{2n\pi}{3} + \frac{\pi}{6}, n \in I$

10. $\theta = n\pi$ or $\theta = \frac{m\pi}{3} \pm \frac{\pi}{9}; n, m \in I$

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Regular Analysis Through Continuous Exercise

Trigonometric Equations: RACE-03

Single Correct Question

1. The number of integral values of a for which the equation $\cos 2x + a \sin x = 2a - 7$ possesses a solution is

(A) 2 (B) 3

(C) 4 (D) 5

MTE159

2. If $x^2 - 4x + 5 - \sin y = 0,\ y \in [0, 2\pi)$, then -

(A) $x = 1, y = 0$ (B) $x = 1, y = \frac{\pi}{2}$

(C) $x = 2, y = 0$ (D) $x = 2, y = \frac{\pi}{2}$

MTE160

3. The number of real solutions of the equation $\sin(e^x) = 5^x + 5^{-x}$ is -

(A) 0 (B) 1

(C) 2 (D) infinitely many

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MTE161

4. If $\sin^3\alpha + \cos^3\beta + 6 \sin\alpha \cdot \cos\beta = 8$, where $\alpha, \beta \in [0, 2\pi]$ then number of ordered pairs (α, β) is equal to

(A) 0 (B) 1

(C) 2 (D) 3

MTE162

Subjective Type Question

5. Solve : $(1 - \sin 2x)(\cos x - \sin x) = 1 - 2\sin^2 x$.

MTE163

6. Solve

(i) $\cos\theta + \sin\theta = \cos 2\theta + \sin 2\theta$

MTE164

(ii) $\cos^2 x + \cos^2 2x + \cos^2 3x = 1$

MTE165

(iii) $\sin^2 n\theta - \sin^2(n-1)\theta = \sin^2\theta$, where n is constant and $n \neq 0, 1$

MTE166

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7. Solve $\sin 3x + \cos 2x = -2$

MTE167

8. Solve $\sqrt{3\sin 5x - \cos^2 x - 3} = 1 - \sin x$

MTE168

9. If $\sin x + \cos x = \sqrt{y + \frac{1}{y}}$, $y > 0$, $x \in [0, \pi]$, then find the least positive value of x satisfying the given condition.

MTE169

10. Find the values of x in the interval $[0, 2\pi]$ for which $4\sin^2 x - 8\sin x + 3 \le 0$.

MTE170

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ANSWER KEY

Question	1	2	3	4
Answer	D	D	A	C

5. $2n\pi + \frac{\pi}{2}, n \in I$ or $2n\pi, n \in I$ or $n\pi + \frac{\pi}{4}, n \in I$

6. (i) $2n\pi, n \in I$ or $\frac{2n\pi}{3} + \frac{\pi}{6}, n \in I$

(ii) $x = (2n+1)\frac{\pi}{4}, n \in I$ or $x = (2n+1)\frac{\pi}{2}, n \in I$ or $x = n\pi \pm \frac{\pi}{6}, n \in I$

(iii) $m\pi, m \in I$ or $\frac{m\pi}{n-1}, m \in I$ or $\left(m + \frac{1}{2}\right)\frac{\pi}{n}, m \in I$

7. $(4p-3)\frac{\pi}{2}, p \in I$ **8.** $2m\pi + \frac{\pi}{2}, m \in I$ **9.** $x = \frac{\pi}{4}$ **10.** $\left[\frac{\pi}{6}, \frac{5\pi}{6}\right]$

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